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Rill formation and evolution caused by upslope inflow and sediment deposition on freshly tilled loose surfaces

Xiaohe Qian [a], Longshan Zhao [a,b,*], Qian Fang [a], Chunhua Fan [a], Ruyi Zi [a], Fayong Fang [a]

[a] College of Forestry, Guizhou University, Guiyang 550025, China
[b] Research Institute of Forestry, Chinese Academy of Forestry, Beijing 100091, China

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ABSTRACT

Rill erosion easily occurs on tilled surfaces because the soil shear resistance is less than the runoff shear stress. However, rill erosion formation and evolution on tilled surfaces under upslope inflow conditions remain unclear. The objective of this study was to investigate the rill formation process and rill erosion amount on tilled surfaces via flow scouring under different inflow rates (4, 6, and 8 L min^{-1}) at a 15° slope. Close-range photogrammetric technology was applied to measure the rill morphology during the experiments. The results suggested that the rill formation process due to inflow could be divided into two distinct stages, namely, before and after runoff reached the downslope end, i.e., stages I and II, respectively. At stage I, runoff washed soil particles along the downslope direction. In this process, due to limited transport capacity of runoff, washed soil particles were deposited at the runoff head and formed a soil mound, which blocked the flow path, after which the runoff direction changed within the 6.7° – 50.6° range with an average moving distance of 0.62 m. As a result, a curved rill and a series of soil mounds were left on the surface. The inflow rate affected rill morphology by influencing the runoff direction change angle and runoff moving distance between the mounds on the surface. Herein, the rill formation process is referred to as downslope-trending erosion by inflow (DTEI). At stage II, runoff reached the downslope end, and a rill channel was formed throughout the slope. Thereafter, DTEI was largely reduced, and headward erosion was strengthened. As a result, the rill morphology quickly changed, such as the rill depth and rill width, which gradually increased with ongoing headward erosion. During DTEI, the rill paths were curved due to sediment deposition, and the sediment deposition conditions varied under the different inflow rates. Therefore, the rill curvature (RC) differed (1.039 ± 0.014). The RC decreased with headward erosion progression at stage II. The total sediment yield (TSY) increased with increasing inflow rate. Under inflow rates ranging from 4 to 6 L min^{-1} and 6–8 L min^{-1}, the TSY increased 2.1–2.4 times. Consequently, DTEI on tilled surfaces significantly affects the initial rill morphology and evolution at the later stage. Hence, on slopes, its role should be considered in rill erosion assessment.

1. Introduction

Rill erosion is an essential soil erosion process that falls between sheet erosion and gully erosion (Poesen et al., 2003). Once a rill is generated, surface flow rapidly converges into concentrated flow with a high erosive force and sediment transport capacity (Wirtz et al., 2013; Hajigholizadeh et al., 2018). This facilitates the development of flow path connectivity on the soil surfaces (Heras et al., 2011; Martín-Moreno et al., 2016). Compared to that during erosion before rill formation, the soil loss is tens of times higher (Di Stefano et al., 2013; Ni et al., 2022). Therefore, rills are considered a primary sediment source and channels for sediment transport on slopes (Mirzaee and Ghorbani-Dashtaki, 2018; Di Stefano et al., 2021).

Previous studies on rill erosion concluded that rills are generated when the runoff shear stress exceeds the soil shear resistance (Knapen and Poesen, 2009; Ou et al., 2021). The rill formation process is as follows: first, knickpoints appear on the slope because of soil compactness differences (Slattery and Bryan, 1992; Mitchell, 2006); then, these knickpoints develop into intermittent rills through headward erosion; finally, the intermittent rills become increasingly connected through further headward erosion, and continuous rills are generated (Brunton and Bryan, 2000; Mahmoodabadi and Cerdà, 2013; Qin et al., 2019). In

* Correspondence to: College of Forestry, Guizhou University, Guiyang 550025, Guizhou, China.
E-mail address: longshanzh@163.com (L. Zhao).

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other words, the rill formation process encompasses four sequential stages: knickpoints, headward erosion, intermittent rills, and continuous rills. With increasing rill erosion development, headward erosion, undercutting erosion and rill sidewall expansion erosion occur (Wells et al., 2010; Bingner et al., 2016; Qin et al., 2019), which constantly alter the rill length (RL), rill depth (RD) and rill width (RW). Hence, the rill morphology varies at different times and at different locations (Bruno et al., 2008; Nicosia et al., 2022). Therefore, most investigators considered knickpoint formation, rill incision and headward erosion as subprocesses of rill formation and commonly adopted the RL, RD, and RW indices to reflect the rill development degree (Kimaro et al., 2008; Di Stefano and Ferro, 2011; Ran et al., 2018).

Rill erosion is affected by rainfall, runoff, soil properties, soil surface microtopography, vegetation coverage, and tillage practices. Rainfall and runoff are the direct drivers, while the remaining factors may indirectly influence the increase or decrease in the impacts of rainfall and runoff (Ilstedt et al., 2007; Wirtz et al., 2013). For example, tillage practices affect rill erosion by altering soil characteristics, surface microtopographic conditions, and vegetation features (Manns et al., 2014; Poesen, 2017; Zhao et al., 2018a, 2018b). Furthermore, on freshly tilled slopes, the surficial soil is greatly loosened, and vegetation is cleared, while the soil compactness of the cultivated layer exhibits the main modification. This plays an essential role in rill formation and development on the soil surfaces. It has been widely suggested that rill initiation and sediment loss resulting from rill erosion are related to soil compactness (Jourgholami et al., 2020; Moinfar et al., 2022). Prats et al. (2019) demonstrated that the sediment yield resulting from splash erosion and sheet erosion on uncompacted bare slopes was half of that on compacted bare slopes, and the sediment yield due to rill erosion on uncompacted bare slopes was nearly four times that on compacted bare slopes. In addition, loose soil on uncompacted slopes can facilitate rainfall infiltration and inhibits runoff formation, which imposes an inhibitory effect on rill generation. However, under a sufficient rainfall intensity, rainfall cannot adequately infiltrate into the soil and rapidly converges into runoff. As a result, rills are quickly formed because the runoff shear stress far exceeds the soil shear resistance within a short time (Evans and Boardman, 2003; Berger et al., 2010; Zhang et al., 2022). Therefore, rill erosion may be more likely to occur on loose soil surfaces under surface runoff generation, while the rill formation process and its influencing factors on loose soil surfaces have rarely been reported.

In previous studies, the sediment yields before and after rill generation have been compared, thereby concluding that the erosion intensity affected by the soil compactness increased when rills emerged on slopes (Di Stefano et al., 2013; Laufer et al., 2016; Dai et al., 2020). Compared to that on compacted slope surfaces, rill erosion is more likely to occur on freshly tilled surfaces or new mounds of abandoned soil with a loose soil layer. However, in previous rill erosion studies, mainly laboratory-simulated rainfall experiments or field observation experiments under natural rainfall conditions have been adopted. Although researchers have investigated the sediment loss attributable to rill erosion through inflow experiments, the rill formation mechanism on loose soil surfaces remains to be determined.

The objective of this study was to explore the formation and morphological development of rills and the rill erosion amount on freshly tilled loose surfaces using a flow scouring experiment under different inflow rates to better understand rill erosion on slopes.

2. Materials and methods

2.1. Soil and soil box

Topsoil (0–20 cm depth) was collected from a slope in the field in Huaxi District (26°28′32″N, 104°42′02″E), Guizhou Province, China. The natural surface soil exhibited a mean bulk density of 1.2 g cm^{-3}. The collected soil was naturally dried at room temperature to an approximately 10.0% mass water content and then passed through a 5-mm sieve to ensure homogeneity. According to the Chinese soil taxonomy, the soil was classified as yellow soil, whose main parent materials include granite, sandstone and shale. Based on the USDA classification, the soil texture was classified as clay loam soil, and the soil composition comprised 31.0% clay, 41.4% sand, and 27.6% silt. The organic matter, total nitrogen and total phosphorus contents were 33.9, 1.8, and 0.6 g kg^{-1}, respectively.

The experiments were conducted in a soil box that was 4.0 m long, 0.5 m wide, and 0.25 m deep (Fig. 1). At the top of the soil box, a panel-type rotor flowmeter (with an error of ± 0.04 L min^{-1}) was used to precisely control the inflow rate, and an energy dissipation pool was divided into two parts via a perforated plate to reduce the inflow energy and adjust the inflow stability (Sun et al., 2018). To achieve a concentrated inflow, the splitter plate at the upslope end of the box was designed as a shallow V-shaped plate, i.e., the middle of the splitter plate was ~5 mm lower than its two sides. At the bottom of the soil box, two rows of holes with a diameter of 5 mm were arranged in parallel along the soil box baseplate at 20-cm intervals to drain the infiltrated water. At the downslope end of the soil box, a V-shaped drainage outlet was installed to collect surface runoff and erosion sediment samples.

2.2. Soil preparation in the box

The slope of the soil box was 15°. The soil box was backfilled with air-dried soil in successive layers with a thickness of 0.05 m each. In this study, the total thickness of backfilled soil in the soil box reached 0.15 m. The mean bulk density and water content in the soil layers were close to those in natural soil layers under field conditions. After the soil box was filled with soil, the soil was ploughed with a hoe, and the soil surface was subsequently evened with a wooden rake. Thereafter, a 30-min prewetting rain process with an intensity of 30 mm h^{-1} was immediately applied to the soil surface 24 h prior to the inflow test using a system of oscillating-nozzle rainfall simulators. The purpose of this prewetting rain process was to restore the soil cohesion after mechanical disturbance during soil preparation and, more importantly, to ensure that the initial soil moisture near the surface was similar during all inflow tests, hence allowing for better comparison of the generated rill erosion associated with the different experimental treatments.

2.3. Inflow test

In this study, three inflow rates (4, 6, and 8 L min^{-1}) were applied. Before each inflow test, we first connected the supply tube of the flowmeter to the water source and then calibrated the flow rate through the flowmeter to meet the test requirements. After the start of the inflow test, a constant inflow into the soil surface was generated from the upslope supply water pool. Thereafter, the produced runoff flowed downwards on the surface. During the experiments, because the soil in the box was relatively loose, the maximum sediment carrying capacity of runoff was quickly reached, and a small mound (or a soil pile such as an alluvial fan) was formed in front of the runoff, leading to runoff direction changes. In other words, the flow direction along the slope changed many times during the experiments. To obtain better experimental data and analyse the results, through observation of the dynamic change in the runoff direction in the trial test, the soil box was divided into three measurement sections from the top of the slope, i.e., upslope (0–1 m), mid-slope (1–2 m), and downslope (2–4 m) sections.

Furthermore, to obtain better rill formation and evolution information during the inflow experiments, each inflow test was divided into six runs (i.e., R1, R2, R3, R4, R5 and R6) according to the number of changes in the runoff direction and the rill morphology. In particular, the first three runs (R1-R3) referred to the conditions before runoff was discharged from the downslope end, while the last three runs (R4-R6) referred to the conditions after runoff was discharged from the downslope end. The inflow supply was immediately stopped at the end of each

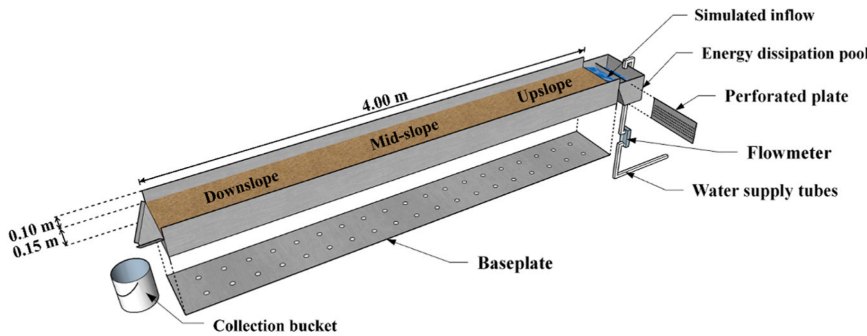

Fig. 1. Diagram of the soil box.

run, and the rill characteristics were immediately measured (the measurement method is described in the next section).

The specific inflow test procedure was as follows: the period from when the inflow entered the top of the upslope to the occurrence of two turns in the runoff direction was defined as the R1 run; the period encompassing the third and fourth turns in the runoff direction was defined as the R2 run; and the period from the end of the fourth turn in the runoff direction to the generated runoff reaching the end of the downslope was defined as the R3 run. Runs R1, R2 and R3 successively occurred in the upslope, mid-slope and downslope parts, respectively. In the R1-R3 runs, the time (T) and distance (D) of each runoff direction change were recorded. After runoff was discharged from the downslope end, two experimental runs with a run time of 10 min were again conducted in the soil box, which were defined as the R4 and R5 runs. In the R4-R5 runs, the inflow rate was consistent with that in the R1-R3 runs, allowing for the investigation of the rill morphology after initial rill generation throughout the slope and comparison of the rill morphology changes to those in the R1-R3 runs. After the R5 run, another experimental run was conducted, which was defined as the R6 run. In the R6 run, the inflow rate was increased by 2 L min^{-1} over the R1-R5 run levels, while the run time was consistent with that of the R4-R5 runs to better understand the effect of inflow rate increase on rill erosion. In the R4-R6 runs, the initial position and displacement of the knickpoints in the rill channel were recorded at 2-min intervals. At the same time, sediment samples were collected at 2-min intervals at the V-shaped drainage outlet and analysed via the drying method (105 °C). In summary, each complete inflow test started with R1 and ended with R6, in which the inflow rate remained unchanged in the R1-R5 runs and was subsequently increased in the R6 run. Two repeat experiments were performed under each inflow rate.

2.4. Rill morphology measurement

Close-range photogrammetric technology was applied to construct high-precision digital elevation models (DEMs) with a spatial resolution up to the millimetre level, which has been widely used in quantitative soil erosion research (Wells et al., 2013; Campbell et al., 2018; Qin et al., 2019; Rodríguez et al., 2022). In each inflow test, surface elevation information of R0 (the initial surface without inflow entry) and R1-R6 was recorded with a Canon EOS 80D(W) (Canon Inc., Kyushu, Japan), which can provide images with 6000 × 4000 pixels. A series of photographs from multiple angles were used for DEM construction to visualize the rill morphological changes. To reduce the influences of the photogrammetric technique on the inflow test, each photogrammetric measurement was finished as quickly as possible in all runs.

2.5. Data analysis

Point cloud data of the slope obtained via close-range photogrammetric technology were extracted and used for DEM construction in Agisoft Metashape Pro 1.7.4 (Agisoft LLC, St. Petersburg, Russia). In each test, seven DEMs were constructed. The detailed steps of DEM construction include importing photos, aligning images, entering control point coordinates, building dense point clouds, and constructing DEMs based on the dense point clouds, which can also be found in the Agisoft Metashape User Manual Professional Edition version 1.8 (https://www.agisoft.com/pdf/metashape-pro_1_8_en.pdf) (accessed on 25 December 2022) and the study of Sun et al. (2021). In the R1-R6 runs, rill morphological parameters (RL, RD, RW, etc.) were extracted from the constructed DEMs by 3D analyst tools in ArcGIS 10.8 (ESRI Inc., California, USA) software, and the rill curvature (RC) can be calculated as follows (Shen et al., 2015):

$$RC = \sum_{i=1}^{n} L_i \Big/ \sum_{i=1}^{n} l_i \quad (1)$$

where RC is the rill curvature, L_i is the length of the i-th rill (m), and l_i is the effective vertical length of the i-th rill (m) (Fig. 2).

To quantify the characteristics of sediment deposition in the R1-R6 runs, the information needed for sediment deposition and rill morphology interpretation was extracted via superposition of the DEMs constructed in the R1-R6 runs and that constructed in the R0 run (Kopyść, 2022). IBM SPSS Statistics 19.0 (IBM Corp., Armonk, N.Y., USA) was used to analyse the total runoff yield (TRY) and the total sediment yield (TSY) differences between the various runs (R4-R6) at the significance level of $\alpha = 0.05$.

3. Results

3.1. Rill formation

The rill formation process is shown in Fig. 3. According to the results of each test, the rill formation process exhibited two distinct stages, namely, the first stage (R1-R3) before runoff was discharged from the downslope end and the last stage (R4-R6) after runoff was discharged

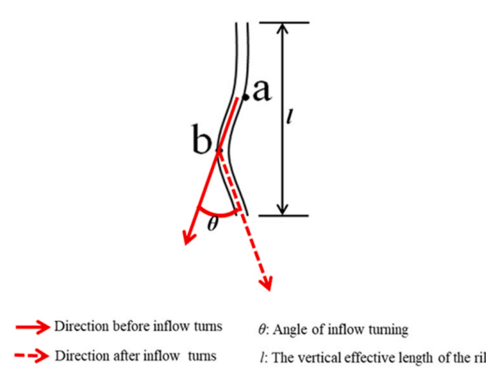

Fig. 2. Schematic diagram of runoff direction change (points a and b are the positions of runoff direction change).

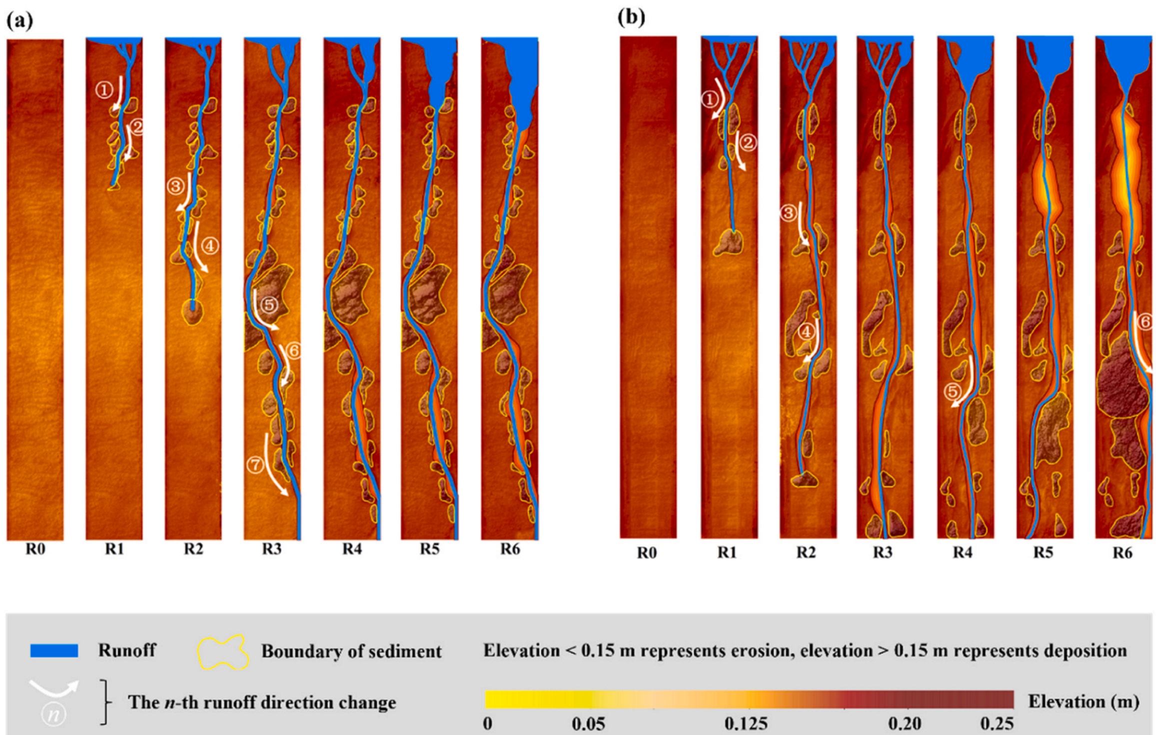

Fig. 3. Visualization of the rill paths and sediment deposition in the R0-R6 runs under the various inflow rates. (a) Inflow rates: 4 L min^{-1} (R1-R5), 6 L min^{-1} (R6); (b) inflow rates: 6 L min^{-1} (R1-R5), 8 L min^{-1} (R6).

from the downslope end.

In the R1-R3 runs, when the inflow entered the slope from the top of the upslope section, incisions were generated across the soil surface, which directed the subsequent inflow. With increasing inflow amount, increasingly more sediment was washed by the generated runoff and transported from upslope to downslope. Furthermore, with increasing runoff sediment load, sediment was deposited in front of the produced runoff, altering the runoff direction (①-⑦).

Sediment deposition mainly occurred in the R1-R3 runs. The occurrence time (T) and distance from the slope top (D) of sediment deposition and the angle of runoff direction change (θ) obviously differed under the various inflow rates (Table 1). For example, under an inflow rate of 4 L min^{-1}, seven instances of sediment deposition occurred before runoff was discharged from the downslope end, and the angle of runoff direction change was 24.8 ± 10.6°. However, under an inflow rate of 6 L min^{-1}, only four instances of sediment deposition occurred, and the angle of runoff direction change was 18.0 ± 6.7°. Compared to those under an inflow rate of 4 L min^{-1}, the total number of changes in runoff was smaller, the distance from the slope top was larger, and the angle of runoff direction change was lower. In addition, the distance required for runoff direction change in the upslope section was shorter than that in the downslope section, and the time required for runoff discharge from the downslope end under an inflow rate of 4 L min^{-1} was 1.42 times that under an inflow rate of 6 L min^{-1}.

In the R4-R6 runs, runoff was discharged from the downslope end, the rill channel cut through the entire slope, and the rill path along the slope was preliminarily determined. Due to the detachment and subsequent transport of the soil at the bottom and sidewall of the rill channel by runoff, the rill channel was continuously deepened and widened. Thus, the rill morphology was greatly affected by the former inflow

Table 1
Time, location and runoff direction of the rills in the different experimental runs.

Runs	Inflow rate: 4 L min^{-1}						Inflow rate: 6 L min^{-1}					
	1			2			1			2		
	T (min)	D (m)	θ	T (min)	D (m)	θ	T (min)	D (m)	θ	T (min)	D (m)	θ
R1	0.62	0.38	44.4°	0.63	0.22	12.5°	0.33	0.55	21.2°	0.67	0.47	15.5°
	1.65	0.70	14.8°	0.73	0.89	12.9°	0.75	1.10	25.8°	1.18	1.24	6.7°
R2	1.87	1.41	20.6°	0.93	1.05	20.1°	1.08	1.63	16.4°	1.62	1.97	25.3°
	2.12	1.84	33.0°	1.42	1.67	29.3°	1.92	3.12	15.1°	2.40	2.52	50.4°
R3	2.52	2.28	33.8°	2.07	1.98	31.6°	2.42	4.00	OF	3.35	4.00	OF
	2.78	3.31	12.4°	2.90	2.62	37.2°	–	–	–	–	–	–
	3.22	3.80	50.6°	4.05	3.66	20.2°	–	–	–	–	–	–
	3.67	4.00	OF	4.50	4.00	OF	–	–	–	–	–	–
	D (m)	n	L (m)	D (m)	n	L (m)	D (m)	n	L (m)	D (m)	n	L (m)
R4	–	–	–	1.3–2.6	3	0.29	1.3–2.6	3	0.56	1.3–2.6	3	0.13
R5	1.3–2.6	4	0.23	1.3–2.6	6	0.26	1.3–2.6	2	0.27	1.3–2.6	4	0.39
	Inflow rate increased to 6 L min^{-1}						Inflow rate increased to 8 L min^{-1}					
R6	1.3–2.6	5	0.26	1.3–2.6	3	0.12	1.3–2.6	2	0.11	–	–	–

T, time of runoff direction change; D, distance from the top of the slope; θ, angle of runoff direction change; n, number of knickpoints; L, average displacement in knickpoint movement; –, no data; OF, runoff discharged from the downslope end. Data for each replication, i.e., 1 and 2, are shown.

conditions in the R1-R3 runs. Under a higher inflow rate, due to the resulting changes in the partial rill morphology, sediment stemming from the upslope part was deposited inside the rill in the downslope part, resulting in runoff direction change. Furthermore, new rill paths, as shown for the R5 and R6 runs in Fig. 3(b), were formed in the downslope part.

When the rills cut across the entire slope, sediment-laden runoff was discharged from the downslope end, and headward erosion emerged in the rills with the knickpoints as the main occurrence locations. In the R4-R6 runs, knickpoints were generated at the bottom of the rill channels at 1.3–2.6 m from the slope top. With headward erosion development, the knickpoints moved upslope with an average displacement (L) ranging from 0.11 to 0.56 m. Under an inflow rate of 4 L min^{-1}, 3–6 knickpoints occurred and moved upslope with an average displacement of 0.92 m. Under an inflow rate of 6 L min^{-1}, 2–4 knickpoints occurred and moved upslope with an average displacement of 1.35 m. Compared to those under an inflow rate of 4 L min^{-1}, fewer knickpoints were generated in the rills, but the knickpoints were displaced across a larger distance under an inflow rate of 6 L min^{-1}. In addition, when the inflow rate was increased to 6 and 8 L min^{-1} from R5 to R6, obvious rill sidewall collapse occurred within the range of 0.6–1.5 m at the slope top, as shown in Fig. 3(b).

3.2. Rill morphology

Variations in the RL and RC are shown in Fig. 4. In the R1-R3 runs, the RL under 4 and 6 L min^{-1} significantly increased, and the RL under 6 L min^{-1} increased more rapidly than that under 4 L min^{-1}. At the end of R3, the maximum RL was obtained. Under inflow rates of 4 and 6 L min^{-1}, the maximum RL values were 4.18 and 4.08 m, respectively. The times needed to reach the maximum RL value under 4 and 6 L min^{-1} were 4.08 and 2.88 min, respectively. Consequently, with increasing inflow rate, the RL first increased in the R1-R3 runs (the RL under 6 L min^{-1} was 1.42 times higher than that under 4 L min^{-1}) and then remained stable in the R4-R6 runs. The higher the inflow rate, the faster the RL increased.

After runoff was discharged from the downslope end, the RC decreased in the R4-R6 runs. In R4 and R5, the RC under 4 L min^{-1} decreased with an average value of 0.0012 per run, and the RC under 6 L min^{-1} decreased with an average value of 0.0031 per run. In R6, the RC values under 6 and 8 L min^{-1} were again reduced by 0.0021 and 0.0057, respectively. The higher the inflow rate, the lower the RC was and the more rapidly the RC decreased.

Fig. 5 shows the changes in the RD at the different slope positions under the various inflow rates. With rill development, the RD increased to varying degrees from R1 to R6. In the R1-R3 runs, the maximum RD values under 4 and 6 L min^{-1} were 0.045 and 0.057 m, respectively. In the R4-R5 runs, the maximum RD values under 4 and 6 L min^{-1} increased by 0.075 and 0.080 m, respectively, while in the R6 run, the maximum RD values under 6 and 8 L min^{-1} increased by 0.109 and 0.079 m, respectively. This demonstrates that the RD in R6 changed the most, followed by that in R4-R5 and R1-R3. With increasing inflow rate, the RD in the upslope part was greater than that in the downslope part.

The characteristics of the RW, namely, the change degree and the slope position of significant RW change, differed in R1-R3 and R4-R6 (Fig. 6). In the R1-R3 runs, the RW changed slightly, and the rill in the upslope part was narrower than that in the downslope part. In the R4-R6 runs, the RW significantly changed, and the rill in the upslope part was wider than that in the downslope part. In addition, in each run from R1 to R5, the rill under 6 L min^{-1} was wider than that under 4 L min^{-1}. In contrast, in R6, the rill under 8 L min^{-1} was wider than that under 6 L min^{-1}. Notably, the higher the inflow rate, the more significantly the RW changed. Therefore, the rills in the upslope part were narrower than those in the downslope part before the rill channel cut through the entire slope, while the relationship between these quantities was gradually reversed afterwards. This change characteristic was more obvious under a higher inflow rate.

Under an inflow rate of 8 L min^{-1}, the rills in the downslope part were filled with a large amount of sediment, even though the elevation of sediment deposition was higher than that of the rill banks. Furthermore, new rill paths were generated on the downslope section, which were narrower than the former rill paths (i.e., R6 in Fig. 6b).

3.3. Runoff and sediment

Fig. 7(a, b, c) shows the variation characteristics of the runoff rate (RR) and TRY in the R4-R6 runs under the various inflow rates. In R4-R5, the RR values under 4 and 6 L min^{-1} fluctuated within the ranges of 19.8–49.6 g s^{-1} and 69.2–121.2 g s^{-1}, respectively. In R6, the RR values under 6 and 8 L min^{-1} fluctuated within the ranges of 71.0–104.2 g s^{-1} and 104.2–217.9 g s^{-1}, respectively. From R4 to R5, the TRY values under 4 and 6 L min^{-1} increased by 7.68 and 23.00 kg, respectively. In R6, the TRY values under 6 and 8 L min^{-1} increased by 34.75 and 27.13 kg, respectively. Moreover, in the same runs, the TRY under 6 L min^{-1} was significantly higher than that under 4 L min^{-1}, while the TRY under 8 L min^{-1} was also significantly higher than that under 6 L min^{-1}. This result suggests that the RR and TRY increased with increasing inflow rate and significantly increased ($P < 0.05$) from R4 to R6 in sequence, and the TRY was proportional to the RR.

Fig. 7(d, e, f) shows the characteristics of the sediment concentration (SC) and TSY in the R4-R6 runs under the various inflow rates. In R4-R6, the SC under the three inflow rates fluctuated within the range of 50–400 g L^{-1} without apparent regularity. Under an inflow rate of 4 L min^{-1}, the TSY values in R4 and R5 were 2.91 and 3.28 kg, respectively. Under an inflow rate of 6 L min^{-1}, the TSY values in R4 and R5 were 12.22 and 12.44 kg, respectively. Notably, the TSY in R4 was similar to that in R5. However, in R6, the TSY values under 6 and 8 L min^{-1} were 7.00 and 27.80 kg, respectively, which are 2.1–2.4 times

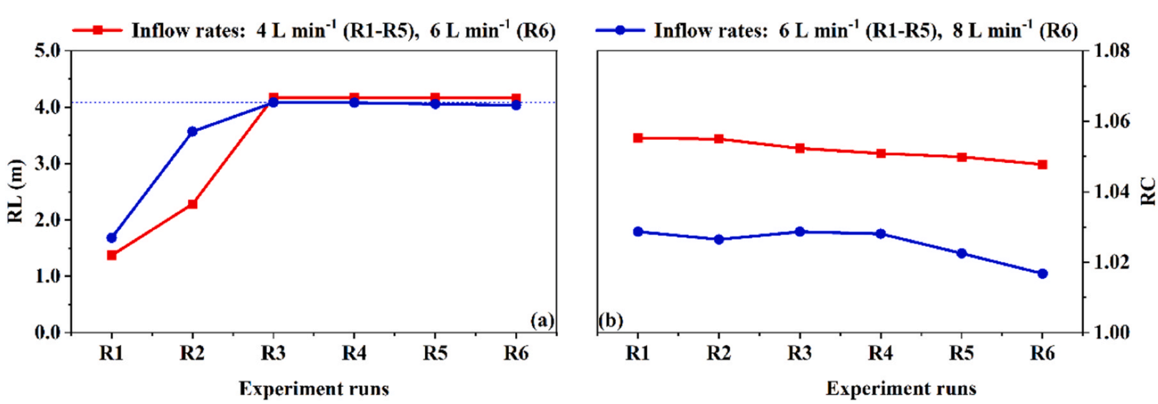

Fig. 4. Variations in the rill length (RL) and rill curvature (RC) in the R1-R6 runs under the various inflow rates. (a) RL; (b) RC.

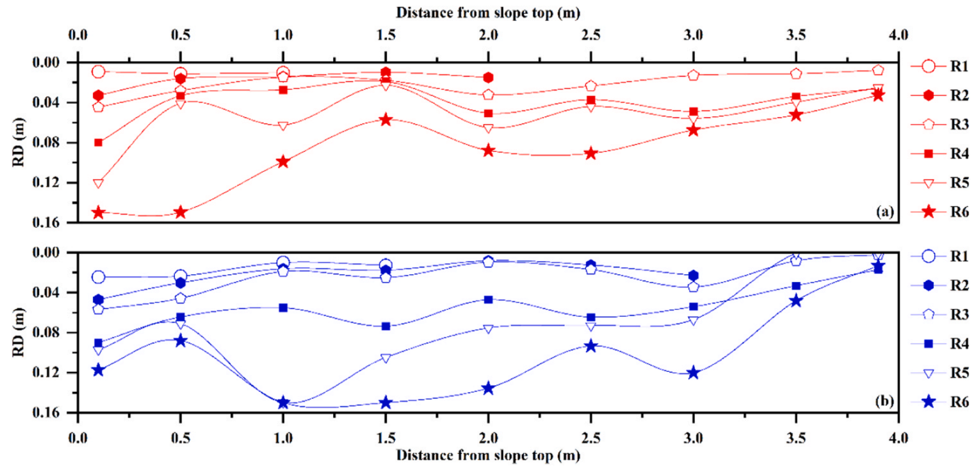

Fig. 5. Variation in the rill depth (RD) at the different positions along the slope in the R1-R6 runs under the various inflow rates. (a) Inflow rates: 4 L min^{-1} (R1-R5), 6 L min^{-1} (R6); (b) inflow rates: 6 L min^{-1} (R1-R5), 8 L min^{-1} (R6).

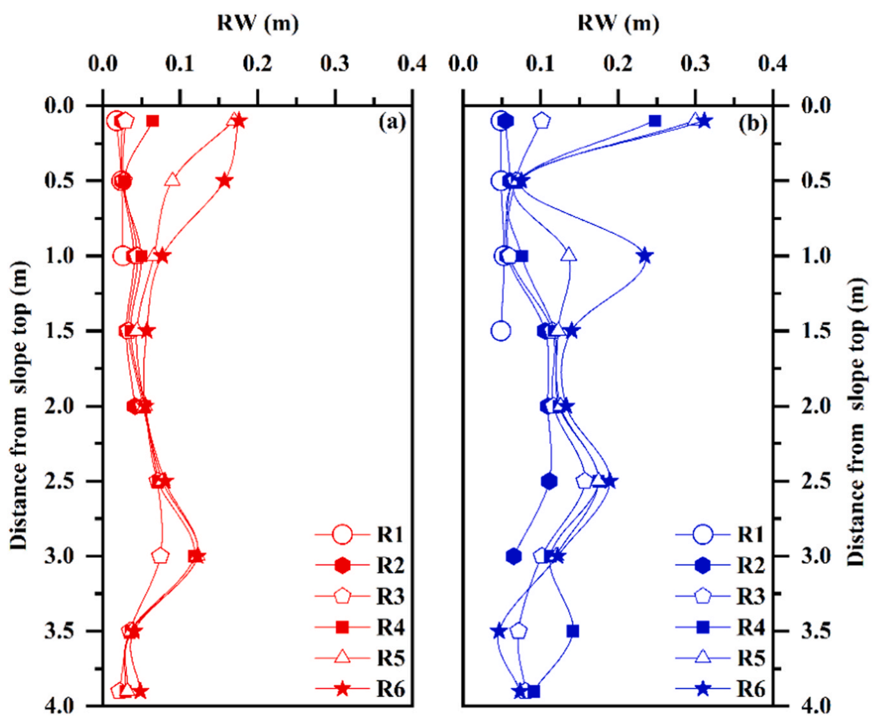

Fig. 6. Variation in the rill width (RW) at the different positions along the slope in the R1-R6 runs under the various inflow rates. (a) Inflow rates: 4 L min^{-1} (R1-R5), 6 L min^{-1} (R6); (b) inflow rates: 6 L min^{-1} (R1-R5), 8 L min^{-1} (R6).

that in R4-R5. Moreover, the TSY under 6 L min^{-1} was significantly higher than that under 4 L min^{-1}, while the TSY under 8 L min^{-1} was also significantly higher than that under 6 L min^{-1} in the same runs ($P < 0.05$). Furthermore, the higher the inflow rate was, the higher the TSY.

4. Discussion

In this study, the effects of inflow on the rill formation process, rill morphology and erosion sediment characteristics on loose soil surfaces were investigated. Based on the experimental observations, we concluded that rill erosion could be divided into two distinct stages (i.e., before and after runoff was discharged from the downslope end, denoted as stages I and II, respectively). At stage I, due to soil surface scouring by the produced runoff, incisions were produced near the top of the upslope. Furthermore, these incisions developed into prototype rills to guide the subsequent inflow. Because the topsoil was unconsolidated, a large amount of soil was carried away by runoff, which was conducive to a rapid increase in the SC. The effect of soil compactness on rill erosion is manifested in two aspects. First, the rill formation process is affected by the infiltration capacity of soil depending on the soil compactness. When soil is compacted, the soil infiltration capacity decreases, which promotes runoff formation and runoff shear stress enhancement, but the soil shear resistance is also significantly increased, thus ultimately generating few rills (Moeyersons et al., 2015; Jourgholami et al., 2020). Conversely, when soil compaction is absent, the increase in the soil infiltration capacity inhibits runoff formation and limits the runoff shear stress, but the soil shear resistance is insufficient, which facilitates rill formation and development. When the SC was increased and further exceeded a threshold equal to the maximum sediment transport capacity

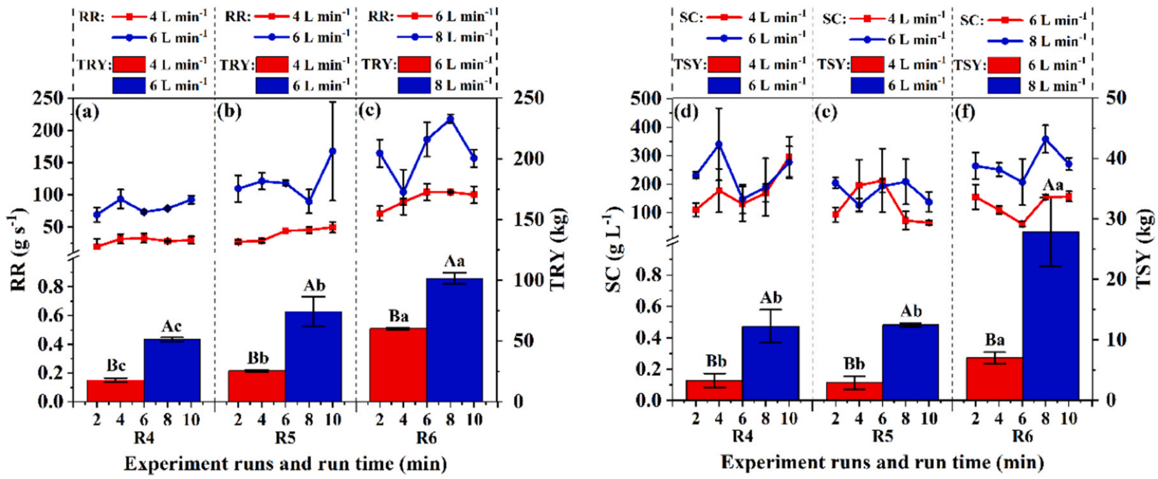

Fig. 7. Runoff rate (RR), total runoff yield (TRY), sediment concentration (SC) and total sediment yield (TSY) in runs R4, R5, and R6 under the various inflow rates (the different lowercase letters indicate significant differences in the TRY, TSY and sediment load among the different stages under the same inflow rate; the different capital letters indicate significant differences in the TRY and TSY among the different inflow rates at the same stage).

of runoff, the transported sediment was deposited at the runoff head, and a soil mound was formed. Because the subsequent runoff was blocked by the soil mound, the runoff direction was adjusted within the range of 6.7°–50.6°. Thereafter, runoff continued to flow downslope along the new direction, and the detached surface soil was carried until sediment deposition occurred at the next position. These subprocesses (i.e., runoff carrying sediment away, sediment deposition, runoff direction change, and runoff continuing to wash sediment away) alternately occurred. Therefore, in this study, a continuous denudation–transport process was observed, referred to as downslope-trending erosion by inflow (DTEI).

Our results indicated that DTEI is one particular formation mechanism of rills on loose surfaces, which is mainly driven by the continuous downslope movement of rill mouths. Therefore, the DTEI mechanism differs from that of headward erosion, which is driven by rill head backward movement. Morphologically, the erosion rills induced by DTEI in the upper part of the slope are larger than those in the lower part (Marzolff and Poesen, 2009; Vanmaercke et al., 2016; Yibeltal et al., 2021). Moreover, DTEI can form a series of soil mounds near the flow path and change the local surface roughness. After rills cut through the entire slope or occur at the stage of headward erosion, these mounds provide an erosion source and hence increase the total amount of soil erosion. Moreover, researchers have previously considered that rill formation is related to the response of the soil shear resistance to the runoff shear stress (Knapen et al., 2007; Hieke and Schmidt, 2013; Ou et al., 2021). When the runoff shear stress exceeds the soil shear resistance, incisions can be first generated in part of the soil surface and then evolve into knickpoints, which provides favourable conditions for rill formation. After the development of knickpoints into intermittent rills via headward erosion, the intermittent rills can develop into continuous rills through further headward erosion (Brunton and Bryan, 2000; Mahmoodabadi and Cerdà, 2013; Zhang et al., 2017). Furthermore, Sun et al. (2020) concluded that rill formation encompassed four subprocesses: knickpoints, headward erosion, intermittent rills, and continuous rills. However, our experimental results demonstrated a different rill formation process caused by different ways of water inflow into the slope. On loose soil surfaces, the soil shear resistance decreases, and the inflow-induced runoff shear stress is enhanced, which provides preliminary conditions for rill formation and development. Under inflow conditions, rills were generated from the top to the end of the slope. Hence, rill formation on slopes can occur in different ways.

According to the test process, the DTEI duration was extremely short (affected by the slope length). In this study, the process only lasted less than 5 min. Thus, DTEI has not received much attention in the past but is a critical rill formation process. In addition, during DTEI, off-site erosion occurred due to the inflow washing away sediment, but sediment was deposited on the slope instead of being discharged. This is also the reason why the actual effect of DTEI has generally been neglected in studies aimed at investigating soil erosion. However, in the DTEI process, due to sediment deposition at the runoff head, the runoff direction changed. The deposited sediment alters the soil surface microtopography, which impacts ground erosion (Berger et al., 2010; Zhao et al., 2014, 2021), while the change in the runoff direction increases the RL on slopes, which inevitably influences subsequent rill erosion (Heras et al., 2011; Di Stefano and Ferro, 2011; Sofia et al., 2017). Our results also confirmed that the angle of runoff direction change (θ) was higher under a lower inflow rate. This occurred because the lower the inflow rate, the more sediment deposition was promoted.

At stage II, headward erosion with knickpoints as carriers occurred in the rills, accompanied by rill sidewall collapse and rill channel deepening. As sediment was discharged from the downslope end, the RC of the curved rills formed at stage I was gradually reduced. Moreover, a large amount of sediment originating from the upslope was directly discharged from the slope through the rill channels, and the RD and RW in the upslope part were therefore greater than those in the downslope part. This phenomenon is similar to the results obtained by Kou et al. (2020) in a gully erosion investigation on the Loess Plateau (i.e., upslope gully erosion was more severe than downslope gully erosion). Consequently, the former rill characteristics attributed to DTEI notably influenced the later rill morphology and sediment yield (Liu et al., 2015; Jiang et al., 2018). In addition, our results indicated that the sediment yield was closely and positively correlated with the inflow rate. Notably, the sediment yield increased with increasing inflow rate.

5. Conclusion

A laboratory experiment was conducted to measure rill erosion on a loose soil surface under different inflow conditions. Two distinct stages of the rill erosion process were identified, i.e., before and after runoff was discharged from the downslope end, which are denoted as stages I and II, respectively. At stage I, the generated runoff drove soil particles downslope, and in this process, the transported sediment was deposited at the runoff head at a certain distance, resulting in a curved rill morphology from upslope to downslope. At stage II, the rills cut through the entire slope, and headward erosion played a major role. Considering that the rill formation and evolution characteristics at stage I completely differed from those of rill erosion induced by headward erosion at stage II, a novel rill formation mode was proposed for loose soil surfaces, namely, DTEI. Because DTEI rapidly occurs on loose soil surfaces, it has been ignored in previous studies. However, this process directly affects rill characteristics and hence greatly influences the soil erosion intensity and process after rills cut through the entire slope. Consequently, DTEI should be considered in future evaluation studies of rill erosion processes on slopes. The obtained results provide a greater understanding of rill formation and evolution on loose soil surfaces.

In this study, a single slope with a 4-m length was adopted, and noteworthy results were obtained. In future studies, it may be important to expand the variable range (such as the slope, slope length, soil water content and presence of plant roots) to better explain the mechanism of DTEI induced by upslope inflow and sediment deposition.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data Availability

Data will be made available on request.

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